



Society/Apartment Management Portal

Project Review 2 — Web Application Development

Future Enhancements & Conclusion

Our roadmap extends beyond the initial launch, focusing on continuous improvement and expanded functionality to deliver even greater value.



Secure Payments

Integrate Razorpay/PayPal for seamless and secure transactions.



Instant Alerts

Implement email and SMS notifications for critical updates and events.



On-the-Go Access

Develop native mobile applications for iOS and Android users.



Data Insights

Provide administrators with a comprehensive analytics dashboard.

This project delivers a robust, scalable foundation for modern apartment management, with a clear roadmap for future growth and innovation.

Problem Statement

Manual Operations

Maintenance fees, complaints, and visitor records managed entirely on paper

Poor Communication

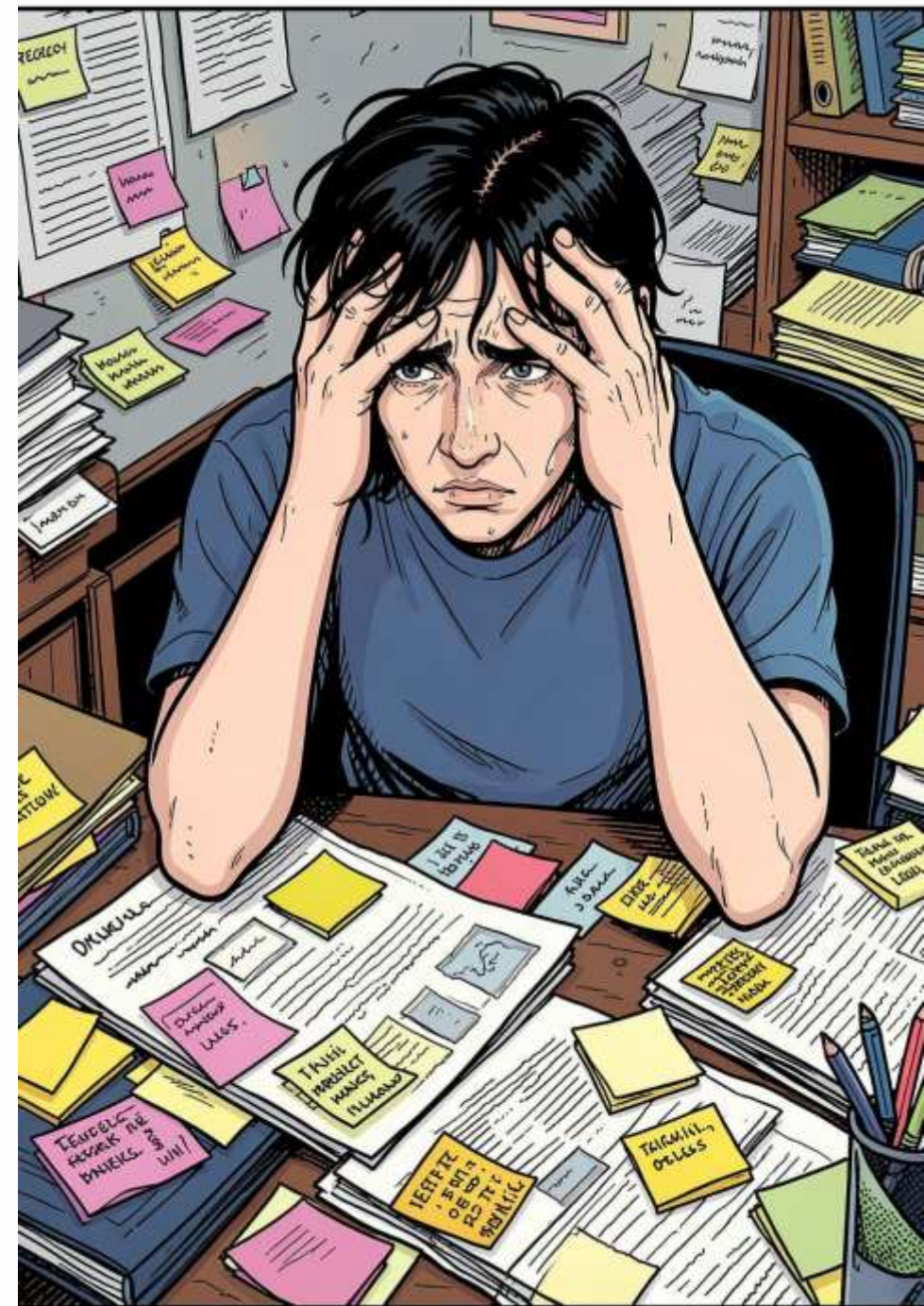
No direct channel between residents and administrators for updates or notices

Payment Tracking Gaps

Difficult to monitor outstanding dues, payment history, and maintenance requests

No Amenity System

Lack of a centralized platform for booking and managing common facilities



Project Objectives



Centralized Web Platform

Unified portal for administrators and residents to manage all apartment operations



Online Payments

Secure digital payment processing for residents — anytime, anywhere



Automated Billing

Automated maintenance bill generation and real-time payment status tracking



Visitor Gate-Pass

Streamlined visitor entry management with digital gate-pass generation

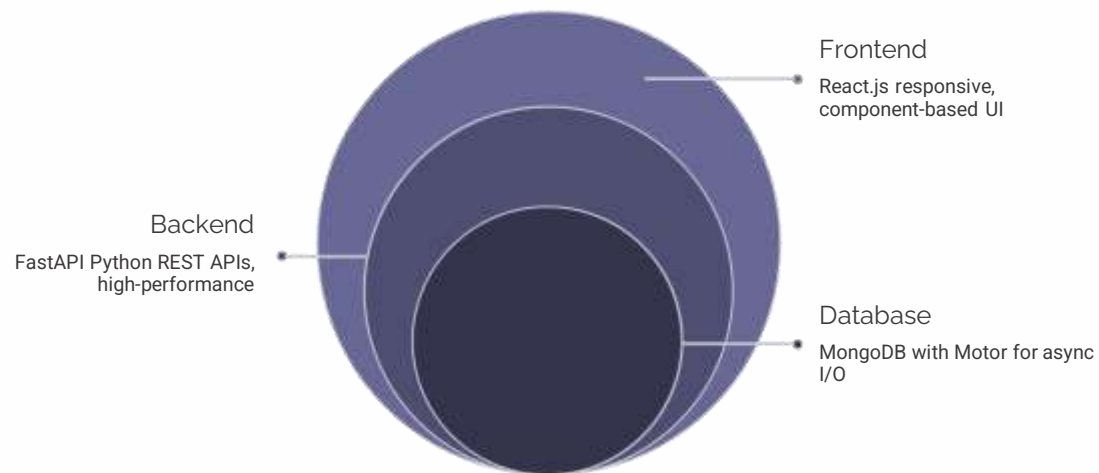
Proposed System Overview



What the Portal Delivers

- **Admin & Resident Portal** — Role-based centralized access for all stakeholders
- **Resident & Flat Management** — Complete flat and occupant information in one place
- **Billing & Payments** — Maintenance bill generation with integrated online payment facility
- **Visitor Gate-Pass** — Digital entry logs and gate-pass issuance for all visitors
- **Amenity Booking** — Reserve common facilities with live availability tracking

System Architecture



Technology Stack

Frontend

React.js – responsive, component-based UI

Backend

FastAPI (Python) – high-performance REST APIs

Database

MongoDB with Motor/PyMongo for async operations

Dev Tools

Postman, Swagger UI, Git version control

Core Modules & Functionalities



Resident Management

Profile creation, flat details, and contact information maintained per resident



Maintenance Billing

Automated bill generation with full payment history and dues tracking



Visitor Management

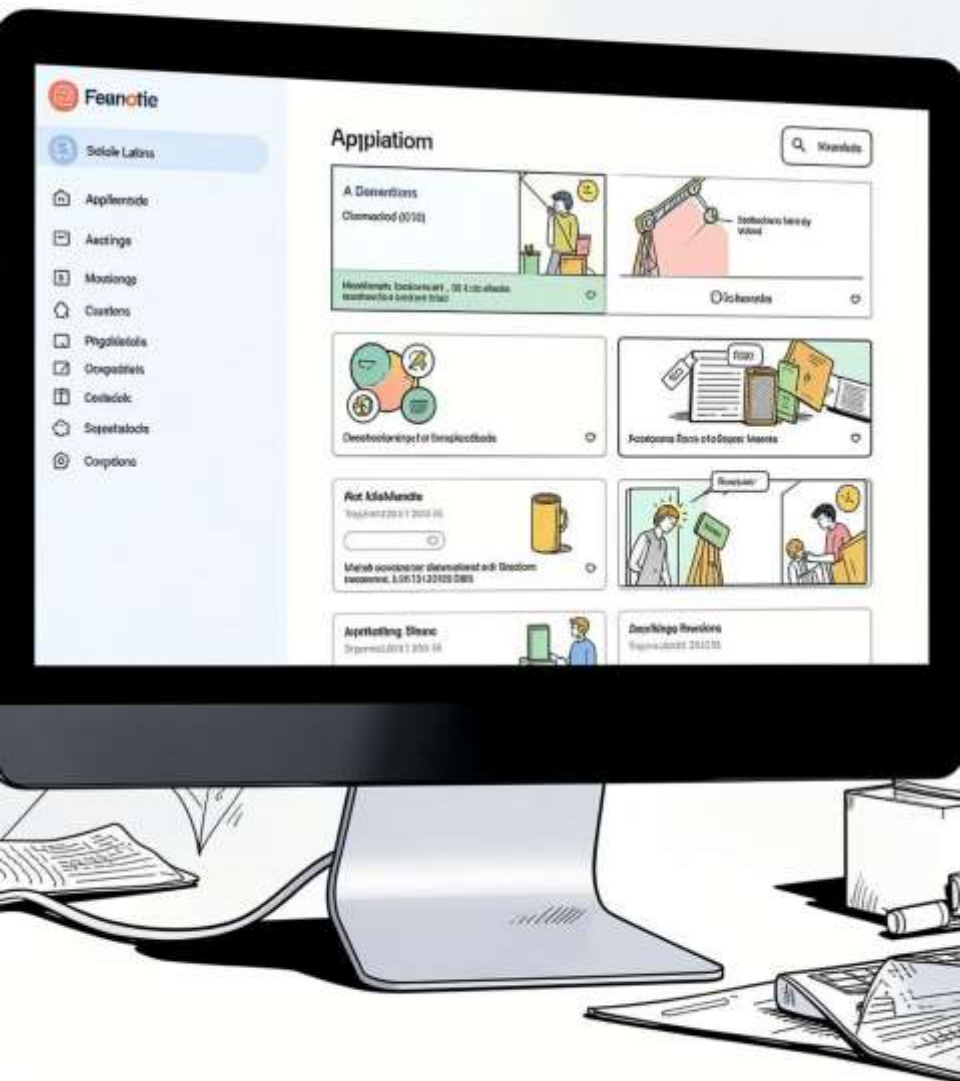
Gate-pass generation, visitor entry logs, and security oversight in real time



Amenity Booking

Reserve common facilities with live availability and scheduling conflict prevention

Frontend Design – Key Pages



1

Home Page

Dashboard overview and quick-access navigation links

2

Login / Register

Secure authentication for both residents and administrators

3

Dashboard

Personalized view of dues, complaints, and active bookings

4

Forms

Complaint submission, payment entry, and amenity booking forms

Frontend Implementation – Screenshots

Key screens developed and tested across resident and admin roles:

Home Page

Layout, hero section, and primary navigation menu

Login & Registration

Authentication forms with validation for residents and admins

Admin Dashboard

Overview panel for managing residents, bills, and complaints

Resident Dashboard

Personalized view of dues, notices, and booking history

Complaint Form

Multi-field complaint submission with category and priority selection

Payment Form

Maintenance bill payment with status confirmation

 Insert live screenshots of each page here during the presentation demo.

Database Design

MongoDB Collections

Residents

Profile, contact, and flat assignment data

Flats

Unit details, floor, and occupancy status

Bills & Payments

Maintenance invoices linked to residents with payment status

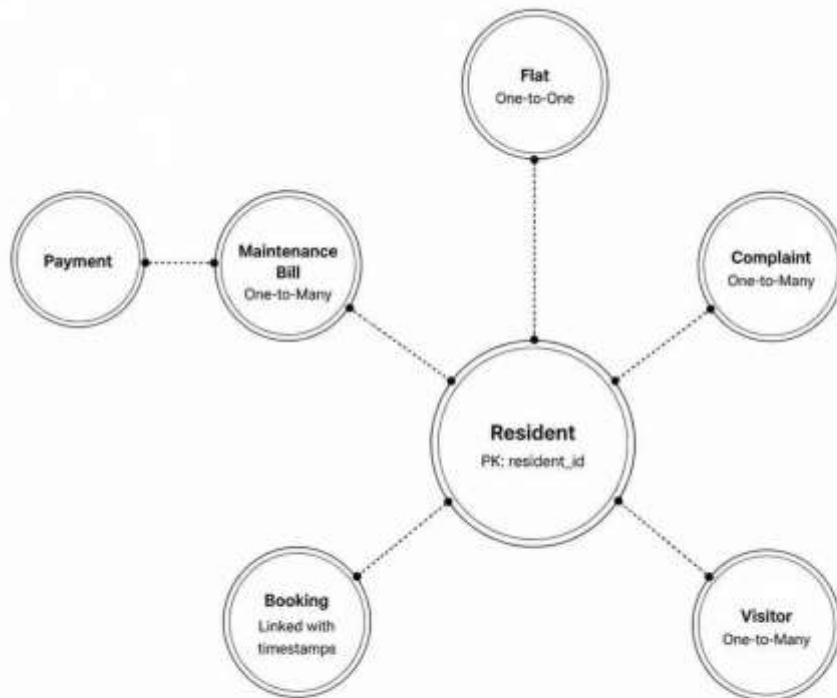
Visitors & Bookings

Gate-pass logs and amenity reservation records with timestamps

Design Decisions

- **Relationships:** Residents linked to Flats, Maintenance Bills, Complaints, and Visitor records
- **Indexing:** Optimized queries on `resident_id`, `flat_number`, and `payment_status` fields
- **Async Driver:** Motor (async PyMongo) for non-blocking database operations with FastAPI
- **Complaints Collection:** Tracks submission date, status, and resolution timestamps

ER Diagram & Database Schema



Key Entity Relationships

- **Resident** → **Flat**: Each resident is assigned to exactly one flat unit
- **Resident** → **Bills**: One resident may have multiple maintenance bills over time
- **Bill** → **Payment**: Each bill tracks one payment record with status and date
- **Resident** → **Complaints**: Complaints logged per resident with category and resolution status
- **Visitor** → **Resident**: Visitor gate-passes linked to the host resident
- **Resident** → **Bookings**: Amenity reservations with timestamps and slot details



Insert actual ER diagram generated from your database schema here during presentation.

Current Progress & Challenges



Completed Work

- Frontend pages (UI/UX)
- Login and Registration flows
- Complaint & Amenity Forms
- Core Navigation Structure



In Progress

- Backend API Endpoints for modules
- Initial Database Optimization & Indexing
- Real-time Notifications for residents



Current Challenge

Optimizing complex MongoDB queries for large datasets and ensuring high performance under concurrent user loads.



Next Steps

- Payment Gateway Integration
- Comprehensive System Testing (Unit, Integration, UAT)
- Deployment Planning & Execution
- User Documentation & Training